



Assessment Cover Sheet

Assessment Title	Project (Group)		
Assessment Type	Uncontrolled	Group	Not must-pass
Due Date	15-Dec-2026	Course Code	IT7103
Course Title	Advanced AI Applications		
Internal Moderator's Name			
External Examiner's Name			

Instructions:

1. This cover sheet must be completed (section in blue below) and attached to your assessment before submission in soft copy.
2. The deadline for this assessment is 15-Dec-2026(11:55 PM).
3. This assessment carries 40% marks, assessing CILO 1 and CILO2.
4. Late submission: marks will be capped at 60%. Two calendar days after the deadline, students will get 0%.
5. The **use of generative AI tools is strictly prohibited**.
6. References consulted (if any) must be properly acknowledged and cited.

Learner ID		Date Submitted	
Learner Name			
Programme Code			
Programme Title			
Lecturer's Name			

By submitting this assessment for marking, I affirm that this assessment is my own work.

Learner Signature

Do not write beyond this line. For assessor use only.

Assessor's Name			
Marking Date		Maks Obtained	

Comments:

- For the given problem statement of the project, the student needs to build the project on Google Collaboratory and upload the downloaded .ipynb file on the submission link, which will be available on Moodle.
- Project submission must be supported with the report where the students must explain the steps required for the completion of the project as per the details given under the “Project Report” section at the end of the project.
- A project report must be in the document format (doc. or docx). Groups are allowed to make submissions only as one Zipped folder containing .ipynb files and report in docx format.
- The Project **Zip** folder for submission must be named with the stream number and only the group leader's student ID. If the stream number is 08 and the group leader student ID is 202316152, the zipped folder must be named "08_202316152".
- Any submission in **RAR** format is **NOT** allowed, and the student will be awarded a zero mark for that submission. Please include a cover page within the report with the full name and student ID of all group members.
- In Google Collaboratory, each of the steps performed for the project must be supported with comments or explanations.
- The submission needs to be uploaded to the course Moodle page.

Project Scenario:

In modern higher education environments, universities generate large volumes of student academic and behavioral data across multiple semesters and courses. As student populations grow, it becomes increasingly difficult for academic staff to manually monitor performance patterns, identify at-risk students, and make timely interventions. To improve academic support and decision-making, educational institutions are turning to machine learning techniques to analyze structured student records and predict academic outcomes.

This project simulates the development of an intelligent academic performance analysis pipeline designed to support real-world educational operations. The goal is to build a machine learning-based solution that examines student data, detects patterns related to achievement, and identifies factors influencing academic success. This can help improve advising efficiency, prioritize support for struggling students, and reduce the time needed to detect poor performance trends.

Dataset Overview:

1. Data Description

This dataset contains academic records for university students enrolled in different courses. It includes student information, course details, study behavior, and performance indicators such as **grade_letter**, **numeric_grade**, and **final_exam**.

The dataset can be used to build predictive models for academic performance, analyze factors affecting student achievement, or classify students based on their results.

2. Available Fields

- **student_id**: Student identifier.
- **semester**: Academic semester.
- **course_code**: Course code.
- **course_name**: Course name.
- **subject_area**: Academic subject area.
- **credits**: Number of credit hours.
- **instructor**: Instructor name.
- **grade_letter**: Final letter grade.
- **numeric_grade**: Numerical grade.
- **prior_gpa**: Previous GPA.
- **study_hours**: Number of study hours.
- **attendance_rate**: Attendance percentage.
- **gender**: Student gender.
- **age**: Student age.
- **year_of_study**: Academic level of the student.
- **scholarship**: Whether the student has a scholarship.
- **extracurricular**: Participation in extracurricular activities.
- **internship**: Internship participation.
- **homework_avg**: Average homework score.
- **final_exam**: Final exam score.

3. Variable Types

The dataset includes a mix of:

- **Categorical variables**: semester, course_code, course_name, subject_area, instructor, grade_letter, gender, year_of_study.
- **Numerical variables**: credits, numeric_grade, prior_gpa, study_hours, attendance_rate, age, homework_avg, final_exam.
- **Binary variables**: scholarship, extracurricular, internship.

4. Analytical Use

This dataset is well suited for **academic performance analysis** and **student outcome prediction**. It can support classification tasks if the target is the letter grade, or regression tasks if the target is the numerical grade or final exam score.

Project Objectives

The project focuses on developing a machine learning pipeline for analyzing student academic performance and predicting educational outcomes. The main objectives are as follows:

1. Basic Data Cleaning and Outlier Handling

Perform basic data cleaning by detecting and handling outliers in the dataset. This step is necessary to improve data quality, reduce noise, and ensure that the records are suitable for further analysis and modeling.

2. Advanced Pre-processing and Feature Engineering

Apply advanced preprocessing techniques to prepare the data for machine learning tasks. This includes creating meaningful features, handling class imbalance for classification tasks, and organizing the dataset in a format suitable for both regression and classification.

For time series-style analysis:

- Sort the data for each student by semester.
- Create lag features such as `final_exam_prev_semester`.
- Generate rolling statistics where applicable.
- Ensure that each student has a continuous sequence of semesters, filling missing semesters with missing values if needed.

3. Correlation and Time Series Analysis

Conduct correlation analysis between the input features and the target variable to better understand relationships within the data.

4. Regression Modeling for Final Exam Prediction

Develop regression models to predict `final_exam` using the processed dataset. This should include both a single model and an ensemble approach. Model performance should be evaluated on the test set using appropriate regression metrics such as RMSE, MAE, and R^2 , along with any other relevant measures.

5. Classification Modeling for Grade Prediction

Develop a multi-class classification model to predict `grade_letter` values such as A, A-, B+, and F. The model should be trained using the processed features while carefully avoiding data leakage. Since `numeric_grade` may act as a direct proxy for `grade_letter`, it should be excluded from the classification feature set unless its effect is being explicitly examined.

If class imbalance is present, appropriate balancing techniques should be applied. The classification model should be evaluated using accuracy, precision, recall, F1-macro, and the confusion matrix.

6. Time Series Forecasting for One Student

For one student with at least six semesters of data, forecast future final_exam scores using time series forecasting methods. The selected student should be filtered from the dataset, and the records should be sorted chronologically by semester.

The analysis should include:

- Exploratory data analysis of the student's time series.
- Selection of a forecasting method time series model depending on the amount and structure of the data.
- Splitting the series into training and testing sets in chronological order.
- Fitting the model on the training portion and forecasting the test period.
- Evaluating the forecasts using RMSE, MAE, MAPE and any other.
- Plotting the actual versus predicted values.

Optional time series cross-validation may also be applied to further assess forecast stability.

7. Ensemble Modeling for Classification and Regression

Develop ensemble models for both classification and regression tasks to improve predictive performance and robustness. For classification, ensemble methods may be used to enhance the prediction of grade_letter. For regression, ensemble methods may be used to improve the prediction of final_exam.

The ensemble approaches should be compared against single models, and their strengths and limitations should be discussed based on the evaluation results.

8. Hyperparameter Optimization

Apply hyperparameter optimization techniques to improve model performance. This should be done using an appropriate search or optimization strategy to identify the best parameter settings for the selected models.

The optimization process should be clearly documented, and the impact of tuning on model performance should be evaluated and discussed for both regression and classification tasks where applicable.

Suggested Technical Directions

The project should demonstrate a well-reasoned approach to data preparation, feature engineering, modeling, and evaluation. Students are expected to clearly explain and justify their methodological choices at each stage of the pipeline:

1. Data Cleaning

Prepare the dataset by detecting and handling outliers, missing values, and inconsistent records. Preprocessing decisions should be explained in relation to their impact on model readiness and analytical reliability.

2. Feature Engineering

Construct relevant features that enhance model learning. This may include lagged variables, rolling statistics, encoded categorical variables, and derived indicators that improve both predictive power and interpretability.

3. Time-based Structure

Organize the data by student and semester to support temporal analysis. Where necessary, create continuous semester sequences and preserve the order of observations to enable meaningful time series-style modeling.

4. Regression and Classification Modeling

Implement and evaluate suitable regression models for predicting `final_exam` or other, and classification models for predicting `grade_letter` or other. The selection of models should be justified based on the task type, dataset structure, and expected outputs. Ensemble methods may also be explored to improve performance.

5. Time Series Forecasting

Select a student with enough semester history and perform forecasting of future values. The forecasting method should be justified according to the data length and temporal behavior, and the results should be evaluated using suitable forecast metrics.

6. Evaluation and Validation

Evaluate the regression task using metrics such as RMSE, MAE, R^2 and any other metrics, and evaluate the classification task using accuracy, precision, recall, F1, and a confusion matrix. If class imbalance exists, it should be addressed and discussed. The final evaluation should include a critical analysis of results, limitations, and the suitability of the chosen methods for the academic prediction problem.

7. Optimization

Use hyperparameter optimization to tune the selected models and improve their predictive performance. The optimization strategy should be explained, and the improvement, if any, should be interpreted carefully.

Project Deliverables:

1. Codebase:

Modular and well-documented implementation provided in **Jupyter Notebook format (ipynb)**. The notebooks should clearly demonstrate the full workflow, including data preprocessing, model training, evaluation, and result visualization. All code should be organized and annotated to ensure clarity and ease of understanding.

2. Project Report:

The report should be well structured and professionally formatted, addressing the following aspects:

- a. A clear explanation of the problem and its relevance within a real-world or domain-specific context.
- b. Statement of project objectives and the AI-based approach adopted to address them.
- c. Description of the data used and steps taken to prepare it for analysis or modeling.
- d. Explanation of the machine learning methods applied, including rationale for choices made.
- e. Evaluation of model performance results, with comparison across approaches.
- f. Interpretation of model behavior, including analysis of influential features or decision-making patterns.
- g. Discussion of practical implications, challenges encountered, limitations of the approach, and possible improvements.
- h. Summary of overall findings and final reflections.